Lab Exercise 1: Installing Python, VS Code / PyCharm, Jupyter Notebook

Python Installation

1. Download Python from the official Python website.
2. Start the Python installer.
3. Enable the option to add Python to the system PATH.
4. Select the installation option and complete the setup.
5. Open Command Prompt/Terminal.
6. Check whether Python is installed by entering:

```
python --version
```

VS Code Setup

1. Download and install Visual Studio Code.
2. Launch VS Code.
3. Open the Extensions section.
4. Search for and install the **Python** extension.
5. Create a folder for Python programs.
6. Open the folder in VS Code.
7. Create a file with the `.py` extension.
8. Write the Python program and run it using the Run button or terminal.

Jupyter Notebook

1. Install Jupyter Notebook using Python/pip.
2. Open Command Prompt or Terminal.
3. Enter:

```
jupyter notebook
```

4. The Jupyter interface will open in the browser.
5. Create a new Python notebook.
6. Enter Python statements in a cell and execute them.

Lab Exercise 2: Write and Execute a Basic Python Program

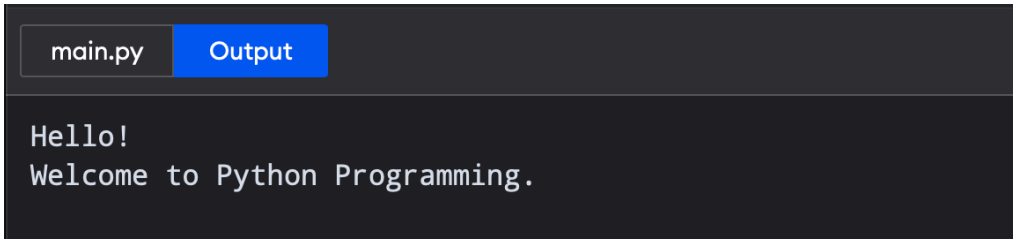
Code :

```
message = "Welcome to Python"

print("Hello!")

print(message + " Programming.”)
```

Output :



Lab Exercise 3: Programs Using Variables, Operators, and Input/ Output

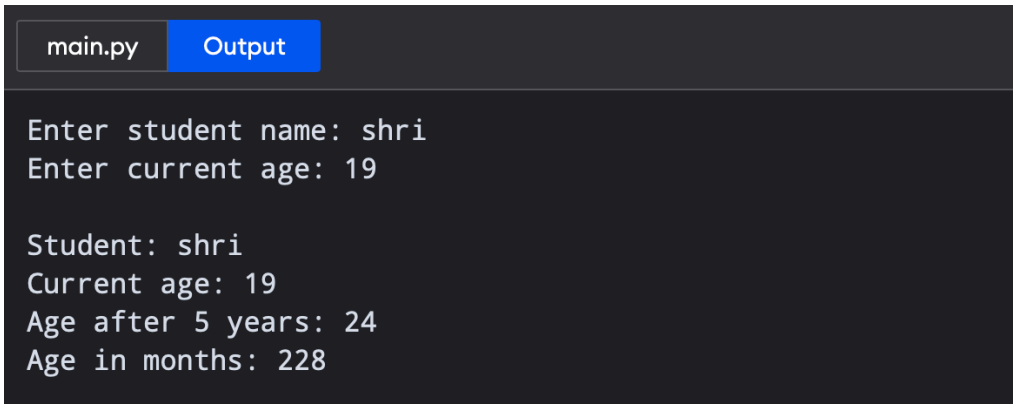
Code :

```
student = input("Enter student name: ")
current_age = int(input("Enter current age: "))

future_age = current_age + 5
age_in_months = current_age * 12

print("\nStudent:", student)
print("Current age:", current_age)
print("Age after 5 years:", future_age)
print("Age in months:", age_in_months)
```

Output :



Lab Exercise 4: Simple Calculator and Area Calculation Programs

Part A — Simple Calculator

Code :

```
print("SIMPLE CALCULATOR")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = int(input("Choose an operation: "))

a = float(input("Enter first value: "))
b = float(input("Enter second value: "))

if choice == 1:
    print("Result =", a + b)

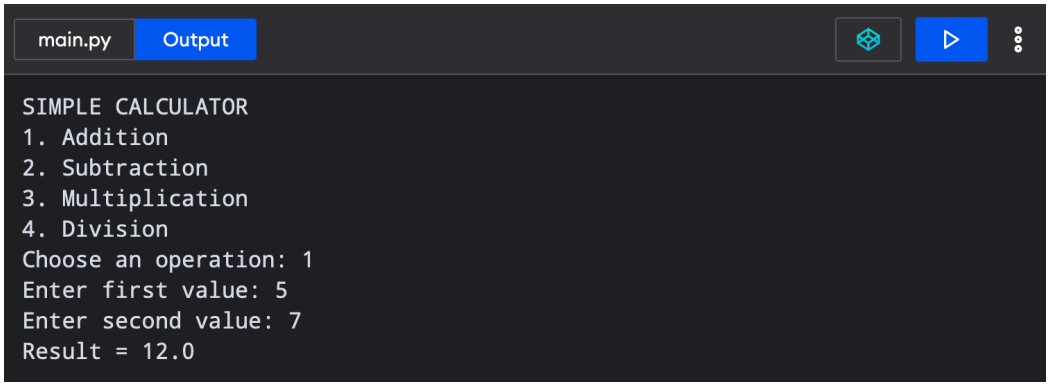
elif choice == 2:
    print("Result =", a - b)

elif choice == 3:
    print("Result =", a * b)

elif choice == 4:
    if b == 0:
        print("Division by zero is not possible")
    else:
        print("Result =", a / b)

else:
    print("Invalid choice”)
```

Output :



Part B — Area Calculation

Code :

```
import math
def rectangle_area(length, width):
    return length * width
def circle_area(radius):
    return math.pi * radius * radius
print("AREA CALCULATOR")
print("1. Rectangle")
print("2. Circle")
option = int(input("Select shape: "))
if option == 1:
    l = float(input("Enter length: "))
    w = float(input("Enter width: "))
    result = rectangle_area(l, w)
    print(f"Rectangle area = {result:.2f}")
elif option == 2:
    r = float(input("Enter radius: "))
    result = circle_area(r)
    print(f"Circle area = {result:.2f}")
else:
    print("Invalid selection”)
```

Output :

